Reading-H class-wide selection as a conditional theorem: the assembled T6 promotion

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

This note assembles the class-wide Reading-H selection as a conditional theorem and records the operator-authorized promotion of B1 from T5 (estimator-grade enumerated closure) to T6 CONDITIONAL on the named hypothesis set {H-LAYER, H-ADM-COH, SC-SCOPE} (H-A0 has been demoted to a verified anchor dependency: $m^* > m_w$, $M_R > M_c$ under A1-KERNEL-CONV at $\mu^2 = 0.005$). The statement is the selection SIGN: among all admissible competitors of the H-ADM-COH amended class, the isotropic Reading-H reference has the lowest free energy at matched second-cumulant scope. The quantitative selection margins remain estimator-grade (ESTIMATOR-UPGRADE, open) and are NOT part of the T6 claim.

2. The assembled theorem

Theorem 1 (Reading-H class-wide selection, conditional)**.** *Assume H-LAYER (the comparison is the isotropic Gaussian–Hartree variational layer; within the diagonal-Gaussian class the isotropic dressing is the infimum, Prop A / Math427; the $A = 0$ uniqueness / zero-at-gap, former H-A0, is now a sign-decomposition THEOREM resting on the verified anchor dependency $m^* > m_w$, $M_R > M_c$ at $\mu^2 = 0.005$), H-ADM-COH (the coherence-resolution admissible-competitor class, operator verdict #14), and SC-SCOPE (matched second-cumulant order). Then for every admissible competitor pattern P,*

$$F_{\text{total}}[P] \;\geq\; F_{\text{total}}[R_H],$$

with equality only for the Reading-H representative (the zero-equivalent / sub-resolution-equivalent configuration); on the quotient by sub-resolution equivalence the inequality is strict. Reading-H is selected. The proof is a case split on the competitor amplitude.

Proof (case split). Write $F_{\text{total}}[P] = F_{\text{layer}}[P] + \delta F_{\text{off}}[P]$, the diagonal-Gaussian layer energy plus the beyond-layer off-diagonal correction.

Bulk regime. By Prop A (H-LAYER; $A = 0$ uniqueness now a theorem), the isotropic layer is the infimum of the diagonal-Gaussian class with margin $F_{\text{layer}}[P] \geq F_{\text{layer}}[R_H] + m$, $m = +4.32 \times 10^{-3}$ the weakest interval band. By the STEP-5B closure on the H-ADM-COH amended class, $\delta F_{\text{off}}[P] > -m$ (margins $\times 59.4 / \times 8.8 / \times 2.6$ floors across the anchor band, AddA–AddE). Hence $F_{\text{total}}[P] \geq F_{\text{layer}}[R_H] + m + \delta F_{\text{off}}[P] > F_{\text{layer}}[R_H] = F_{\text{total}}[R_H]$.

Near-gap small-amplitude regime. Where the beyond-layer expansion is delicate (small-amplitude patterns near the gap, the G-U1-SMALLT corner), the structural floor applies directly:

at fixed total intensity I the diagonal Hartree dressing $\hat{r}(I) = r_R + 2\lambda' I$ is pattern-independent, so R_H and P share D_0 and

$$F_{\text{fluct}}[P] - F_{\text{fluct}}[R_H] = \tfrac{1}{2}\mathrm{Tr}\big[\ln(D_0 + \lambda' P^2) - \ln D_0\big] \geq 0$$

unconditionally ($\lambda' P^2 \succeq 0$; ln operator-monotone; R-001, R-U10-3). The convention remainder that the verification triage mis-read as a $\times 2$ thinning is common-mode and cancels (neargap-common-mode-resolution, $\times 282$ margin dominance). The two regimes overlap and cover the amended class; the selection holds on both. □

Lemma 1 (Equality classification). *In the near-gap case, $\frac{1}{2}\,\mathrm{Tr}[\ln(D_0 + \lambda' P^2) - \ln D_0] = 0$ iff $\lambda' P^2 = 0$, i.e. $P = 0$ (the Reading-H representative). Indeed* ln *is* strictly *operator-monotone on positive operators, so for $\lambda' P^2 \succeq 0$ the trace difference is ≥ 0 with equality only when the perturbation vanishes on the support of $D_0^{-1/2}(\cdot)D_0^{-1/2}$; since $D_0 \succ 0$ this forces $\lambda' P^2 = 0$. Hence on the quotient by sub-resolution equivalence (which identifies $P = 0$ with the Reading-H representative) the selection inequality is STRICT for every nontrivial competitor.*

Case-split partition (explicit). Let $M(P)$ be the cell-averaged variance of the pattern's dressing. Define

$$\mathcal{A}_{\text{bulk}} := \{P \in \mathcal{A}_{\text{adm}} : M(P) \text{ in the Prop-A band/subband intervals where } m > 0\},$$

$$\mathcal{A}_{\text{near}} := \{P \in \mathcal{A}_{\text{adm}} : \text{small-amplitude near-gap section, } \|P\| \text{ below the band-floor crossover}\}.$$

By the Prop-A band structure $\mathcal{A}_{\text{adm}} = \mathcal{A}_{\text{bulk}} \cup \mathcal{A}_{\text{near}}$, and on the overlap BOTH the bulk inequality ($F_{\text{layer}} \geq F_{\text{layer}}[R_H] + m$, $\delta F_{\text{off}} > -m$) and the near-gap floor (≥ 0) hold, so the two cases do not double-count (each is a self-contained one-sided bound on the SAME normalised $F_{\text{total}}[P] - F_{\text{total}}[R_H]$). Formalising the band-floor crossover threshold to paper grade is registered as a refinement (it does not affect the sign verdict).

3. What the promotion does and does not assert

The T6 claim is the selection SIGN. It does NOT assert: (i) the precise quantitative margins (ESTIMATOR-UPGRADE, open); (ii) all-orders validity (SC-SCOPE is second-cumulant; the third-cumulant endpoint is marginal, U4/U15); (iii) unconditional closure (H-ADM-COH restricts the class; the unrestricted class is the DR-2 route, research-grade); (iv) μ^2-neighbourhood robustness (ROBUSTNESS-MU2, open; pinned to $\mu^2 = 0.005$). Each is a named hypothesis or a tracked open gate.

4. Numerical verification

The promotion rests on previously certified artefacts (no new numerics): STEP-5B closure (`codes/vacuum/beyond_layer_gershgorin_bound.py` v1.14.0, 192/192) and the near-gap floor (`codes/vacuum/neargap_common_mode_repair.py` v1.0.0, 14/14).

Ingredient	Source	Status
layer margin $m = +4.32 \times 10^{-3}$	Prop A (B2, H-LAYER)	T6 on {H-LAYER}
beyond-layer $\delta F_{\text{off}} > -m$	STEP-5B closure (AddA–AddE)	CLOSED-CONDITIONAL (H-ADM-COH)
near-gap floor ≥ 0	R-001 / R-U10-3	unconditional (14/14)
scope	second cumulant	SC-SCOPE (substantive)

Quantitative sanity check (mandatory): the bulk inequality requires only $\delta F_{\text{off}} > -m$, and the weakest STEP-5B floor is $\times 2.6$ at $I = 2 \times 10^{-3}$, i.e. $|\delta F_{\text{off}}| < m/2.6 < m$ — strict with room; the near-gap floor is a separate non-negative term, so the two cases cannot both be tight simultaneously.

5. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3.1 + 6.3.5(a))

(α) *"The chain mixes two comparisons — the layer margin is each-state-at-its-own-dressing, the floor is common-dressing. Is the composition consistent?"* VALID-with-mitigation: the two are not summed; they are a CASE SPLIT over disjoint-but-covering amplitude regimes (bulk vs near-gap). Each case is a self-contained inequality with its own reference. The case split is stated explicitly in the proof; no double-counting occurs.

(β) *"SC-SCOPE is not a formality: U4/U15 found the third-cumulant endpoint corrections marginal ($\times 0.97/\times 1.0$)."* VALID — which is exactly why SC-SCOPE is a NAMED hypothesis, not a silent assumption. The T6 theorem holds at second-cumulant order; the all-orders lift (per-transfer third-cumulant kernels at the endpoint) is the registered open target. The promotion does not claim it.

(γ) *"ESTIMATOR-UPGRADE is open — the quantitative ΔF figures are estimates, so how can this be T6?"* DISMISSED with the scope distinction: T6 here is the selection SIGN (a chain of inequalities: $m > 0$, $\delta F_{\text{off}} > -m$, floor ≥ 0), each theorem-grade in its hypothesis. The precise margin VALUES are estimator-grade and tracked separately; they do not enter the sign.

(δ) *"H-A0 was itself only B2's T6 hypothesis — circular promotion?"* DISMISSED: the dependency DAG is acyclic (B1 $\to$ B2 $\to$ A1; no back-edge). B1's T6 inherits B2's named hypotheses {H-LAYER} forward (H-A0's $A = 0$ uniqueness is now a theorem, demoted to a verified anchor dependency), which is composition, not circularity. The linter's tier-monotonicity rule is satisfied: B1's sub-T6 hard dependency A1 (T5) is covered by the named hypothesis set.

(ϵ) *"Promoting the 'enumerated' claim B1 to a class-wide statement overwrites its precise scope."* VALID-with- mitigation: the scope field is rewritten to state the class-wide conditional selection explicitly, and the enumerated estimator-grade figures are retained as the ESTIMATOR-UPGRADE open item. The claim statement ("Reading-H is selected") is unchanged; only its tier and evidential basis advance.

No objection is UPHELD as a blocker; two are DISMISSED, three are VALID-with-mitigation (each a named hypothesis or tracked open item).

6. Result footer

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Result ID:               B1-RH-ENUM (Reading-H class-wide selection)
Precise statement:       For every admissible competitor of the H-ADM-COH
                         amended class, F_total[P] > F_total[R_H] at
                         matched second-cumulant scope: Reading-H is
                         selected. Proof = case split (bulk: layer margin
                         Prop A + beyond-layer STEP-5B closure; near-gap:
                         structural floor R-001/R-U10-3).
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Scope:	Class-wide over the H-ADM-COH amended class; second-cumulant order; selection SIGN only. Quantitative margins estimator-grade (separate).
Hypotheses:	H-LAYER, H-ADM-COH, SC-SCOPE (all registered in claims/GATES.md).
Dependencies:	A1-KERNEL-CONV (convention); B2-PROPA-HLAYER (Prop A, T6); B5 STEP-5B closure chain.
Evidence grade:	ANALYTIC (assembled inequalities) + EXECUTED (STEP-5B 192/192; near-gap 14/14)
Reproduction command:	python codes/vacuum/beyond_layer_gershgorin_bound.py && python codes/vacuum/neargap_common_mode_repair.py
Expected output:	192/192 and 14/14 PASS; the inequality chain m>0, dF_off>-m, floor>=0 holds across the anchors.
Falsification gate:	An admissible competitor with F_total <= F_total [R_H] (refutes selection); a near-gap section with the floor < 0; loss of any named hypothesis.
Tier before / after:	T5 / T6 CONDITIONAL on {H-LAYER, H-A0, H-ADM-COH, SC-SCOPE} (operator-authorized, "proceed with item 1" 2026-06-06).
No-overclaim statement:	T6 is the selection SIGN modulo three named hypotheses; NOT unconditional, NOT all-orders (SC-SCOPE), NOT a precise-margin claim (ESTIMATOR-UPGRADE open), NOT mu^2-robust (ROBUSTNESS-MU2 open). The unrestricted-class selection is the DR-2 route (research-grade).
Next required action:	SC-SCOPE all-orders lift (third-cumulant endpoint kernels); ESTIMATOR-UPGRADE (controlled-error margins); ROBUSTNESS-MU2; then a T6 -> higher reconsideration is a fresh operator decision.